

Bioinformatics Automation Script Report

Ali Ali
dept. of genetics biotechnology
program faculty of Agricultural
Sohag University
Sohag, Egypt
A.Mahmoud2362@nu.edu.eg

Esraa Ibrahim
Botany Dept., Faculty of science,
Tanta University,
Tanta, Egypt
esraa.ibrahim9291@gmail.com

Mai Hussien
Zoology dept. faculty of Science,
Suez University,
Suez, Egypt.
mai.ibrahim@suezuni.edu.eg

Rofaida Abdullah
Faculty of Pharmacy
Zagazig University
Zagazig, Egypt
email address or ORCID

Abstract— This research presents an automation script for bioinformatics tasks, integrating database retrieval, sequence alignment, multiple sequence alignment (MSA), and phylogenetic tree construction. The script is designed to be user-friendly, providing prompts for inputs and options, and utilizes various bioinformatics tools to achieve its functionalities.

Keywords— *Bash Script, Alignment, Blast, Phylogenetic tree*

I. INTRODUCTION

The objective of this project is to develop a bash script that automates various bioinformatics tasks, including database retrieval, sequence alignment, multiple sequence alignment (MSA), and phylogenetic tree construction. The script aims to streamline the bioinformatics analysis workflow by providing a user-friendly interface and integrating commonly used tools and algorithms.

Bioinformatics involves the application of computer science and information technology to the field of molecular biology. BLAST is a widely used tool for comparing biological sequences, providing valuable insights into sequence similarity and evolutionary relationships.

Bioinformatics plays a crucial role in molecular biology by facilitating the analysis of biological data using computational tools. BLAST is one such tool that allows researchers to compare biological sequences and infer functional and evolutionary relationships.

Bioinformatics involves the use of computational tools and techniques to analyze biological data. Automation scripts can significantly enhance the efficiency of bioinformatics tasks, enabling researchers to process large datasets quickly and accurately. This research focuses on developing an automation script that integrates database retrieval, sequence alignment, MSA,

and phylogenetic tree construction, providing a comprehensive solution for bioinformatics analysis.

II. METHODS

The automation script is implemented in Bash and utilizes several bioinformatics tools, including BLAST for database retrieval and sequence alignment, MUSCLE and MAFFT for MSA, and PhyML for phylogenetic tree construction. The script prompts the user for inputs such as database selection, sequence files, and tool options, ensuring flexibility and ease of use.

Step 1: Database Retrieval

In this step, the user selects the desired database and specifies the search criteria to retrieve relevant sequences.

Step 2: Sequence Alignment

The user inputs sequences and chooses alignment parameters to perform sequence alignment using the selected tool.



Fig. 1. ScreenShot Shows a small part of the used code for input sequences choices.

Step 3: Multiple Sequence Alignment (MSA)

The MSA tool interface allows users to perform multiple sequence alignment and visualize the results for further analysis.

Fig. 2. ScreenShot Shows a small part of the used code for MSA.

Users can generate phylogenetic trees from aligned sequences using the phylogenetic tree construction tool.

Fig. 3. ScreenShot Shows a small part of the used code for phylogenetic tree construction tool.

The automation script successfully integrates the selected bioinformatics tools, allowing users to perform a range of tasks from database retrieval to phylogenetic tree construction. The script's modular design and user-friendly interface make it accessible to researchers with varying levels of bioinformatics expertise.

Database Retrieval

Sequence Alignment

Figure 4 illustrates the sequence alignment process, where users can input sequences and choose alignment parameters.

Figure 4 displays the MSA tool interface, allowing users to perform multiple sequence alignment and visualize the results.

Figure 4 depicts the phylogenetic tree construction tool, enabling users to generate phylogenetic trees from aligned sequences.

Design and Implementation

The script is designed to be modular and well-documented, following best practices for bash scripting. Each bioinformatics task is implemented as a separate function, allowing for easy comprehension and future modifications. The script prompts the user for inputs and options, executes the specified tasks using appropriate tools and algorithms, and provides clear output.

To use the script:

1. Ensure that the required bioinformatics tools are installed (raxml, fasttree, mafft, clustalo, muscle, bash, curl, jq, ncbi-blast+, primer3).
2. Run the script in a terminal by executing the command `bash bioinformatics_automation.sh`.
3. Follow the on-screen prompts to specify the desired database, input sequences, and tasks.
4. Provide input sequences in FASTA format when prompted.
5. The script will execute the selected tasks and generate the desired output.

- User selects NCBI database for retrieval.
- User provides input sequences in FASTA format.
- Script performs sequence alignment using BLAST.
- Script conducts MSA using Clustal Omega.
- Script constructs a phylogenetic tree using FastTree.
- Phylogenetic tree is outputted in Newick format.

Functionalities, Limitations, and Potential Applications

1) *Functionalities:*

- Database retrieval, Users can specify a bioinformatics database and retrieve relevant sequences based on search criteria.
- Sequence alignment: Supports alignment of input sequences using various alignment tools.
- MSA: Performs multiple sequence alignment using popular MSA tools.
- Phylogenetic tree construction: Generates phylogenetic trees based on aligned sequences.

2) Limitations:

- Dependency on external tools and databases may require proper setup and installation.
- Limited to the availability and compatibility of bioinformatics databases and tools.

3) Potential Applications:

- Research: Facilitates analysis of biological sequences for research purposes.
- -Education: Provides a hands-on learning experience for students studying bioinformatics.
- Industry: Supports bioinformatics analysis in pharmaceutical, biotechnology, and healthcare industries.

Challenges and Future Improvements

Challenges encountered during development include ensuring compatibility across different environments and handling errors gracefully. Future improvements could include adding support for additional databases and tools, optimizing performance, and enhancing user interface and error-handling capabilities. The automation script provides a valuable tool for bioinformatics researchers, enabling efficient and accurate analysis of biological data. Future improvements could include additional functionalities, such as support for different alignment algorithms and tree visualization tools, further enhancing the script's utility.

IV. CONCLUSION

The bioinformatics automation script provides a valuable tool for researchers, educators, and industry

professionals to conduct bioinformatics analyses efficiently and effectively. By integrating different bioinformatics tasks into a single, user-friendly tool, the script enhances productivity and enables reproducible research in the field of bioinformatics. Overall, the bioinformatics automation script enhances the efficiency of conducting searches by streamlining the process. Future improvements could include error handling for invalid inputs and support for additional functionalities.

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CONTRIBUTIONS

- Script design and implementation: [Ali Mahmoud Ali, Esraa Mohamed Ibrahim]
- Report writing: [Mai Ibrahim Hussien]
- Presentation generator: [Rofaida Belal Abdullah]